

SQL SERVER 2016

chapter 05: how to code summary queries

INSTRUCTOR: MD. HABIBUL HAQ

domain.habib@gmail.com

Q1. Which one is a scalar function?

- A. A function that operates on a single value and returns a single value
- B. A function that operates on a series of values and returns a single value
- C. A function that always returns the same value for a given set of arguments
- D. A function that takes no parameter

Answer: A

Q2. Which one is an aggregate function?

- A. A function that operates on a single value and returns a single value
- B. A function that operates on a series of values and returns a single value
- C. A function that always returns the same value for a given set of arguments
- D. A function that takes no parameter

Answer: B

Q3. Why aggregate functions are column functions?

- A. Because they can be used in column definitions in tables
- B. Because they typically operate on the values in a column
- C. Because they typically add extra columns in the result set
- D. All of the above

Answer: B

Q4. Which queries are referred to as summary queries?

- A. A query that contains one or more user-defined functions
- B. A query that returns a single scalar value
- C. A query that contains GROUP BY clause
- D. A query that contains one or more aggregate functions

Answer: D

Q5. Which aggregate function does not always require an expression to operate on?

- A. AVG
- B. MAX
- C. MIN
- D. COUNT

Answer: D

Q6. Aggregate functions always exclude NULL values.

- A. True
- B. False

Answer: A

Q7. Which one returns number of rows in the base table?

- A. COUNT (ALL expression)
- B. COUNT (DISTINCT expression)
- C. COUNT ()
- D. COUNT (*)

Answer: D

Q8. On which functions ALL or DISTINCT has no effect?

- A. SUM and AVG

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- B. SUM and COUNT
 - C. MIN and MAX
 - D. COUNT and AVG

Answer: C

Q9. Which functions accept expression that results in only a numeric value?

- A. SUM and AVG
- B. SUM and COUNT
- C. MIN and MAX
- D. COUNT and AVG

Answer: A

Q10. By default, all aggregate functions exclude duplicate values.

- A. True
- B. False

Answer: B

[By default, all values are included in the calculation regardless of whether they are duplicated.

Ref. Page: 161]

Q11. If your query has non-aggregate columns with aggregate functions, what should you do to the query to work? [There may be multiple answers]

- A. Specify all non-aggregate columns in ORDER BY clause **a**
- B. Specify all non-aggregate columns in GROUP BY clause
- C. Include OVER clause with each aggregate function
- D. Include COMPUTE BY clause in your query

Answer: B, C

Q12. Which query or queries will work?

- A. Select Country, Count(CustomerId) AS 'No. of customers' FROM Customers
- B. Select Country, Count(CustomerId) AS 'No. of customers' FROM Customers ORDER BY Country
- C. Select Country, Count(CustomerId) AS 'No. of customers' FROM Customers GROUP BY Country
- D. Select Country, Count(CustomerId) AS 'No. of customers' FROM Customers SORT BY Country

Answer: C

Q13. Which query or queries will work?

- A. Select Country, Count(CustomerId) AS 'No. of customers' FROM Customers GROUP BY Country WHERE Country = 'Germany'
- B. Select Country, Count(CustomerId) AS 'No. of customers' FROM Customers WHERE Country = 'Germany' GROUP BY Country
- C. Select Country, Count(CustomerId) AS 'No. of customers' FROM Customers GROUP BY Country HAVING Country = 'Germany'
- D. Select Country, Count(CustomerId) AS 'No. of customers' FROM Customers HAVING Country = 'Germany' GROUP BY Country

Answer: B, C [Having Clause is used with GROUP BY and it must be after GROUP BY clause. WHERE clause can be used but before GROUP BY]

Q14. Which query or queries will work?

- A. Select City, Count(CustomerId) AS 'No. of customers' FROM Customers GROUP BY City WHERE Country = 'Germany'
- B. Select City, Count(CustomerId) AS 'No. of customers' FROM Customers WHERE Country = 'Germany' GROUP BY City

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- C. Select City, Count(CustomerId) AS 'No. of customers' FROM Customers GROUP BY Country, City HAVING Country = 'Germany'
 - D. Select City, Count(CustomerId) AS 'No. of customers' FROM Customers HAVING Country = 'Germany' GROUP BY City

Answer: B

[Having can include columns that are in group by list. Where can you include any column. Where clause must be before group by]

Q15. Which query is correct?

- A. SELECT Country, COUNT (*) FROM Customers WHERE COUNT (*) >5 GROUP BY Country
- B. SELECT Country, COUNT (*) FROM Customers GROUP BY Country HAVING COUNT (*) >5
- C. SELECT Country, COUNT (*) FROM Customers HAVING COUNT (*) >5 GROUP BY Country
- D. SELECT Country, COUNT (*) FROM Customers GROUP BY Country WHERE COUNT (*) >5

Answer: B

[If filter has aggregate it must be with HAVING Clause]

Q16. Which is not extension operator of SQL Server for summarizing data?

- A. ROLLUP
- B. CUBE
- C. OVER
- D. EXCEPT

Answer: D

[4 operators: ROLLUP, CUBE, GROUPING SETS and OVER]

Ref. Page: 172]

Q17. Which of the following is **not** true for the ROLLUP operator?

- A. It adds a summary row for each specified group
- B. It adds a summary row at the end of the result set that summarizes the entire result set
- C. If you use ROLLUP, you can use DISTINCT keyword in any of the aggregate functions used in the query
- D. If you use ROLLUP, you can add ORDER BY clause in your query

Answer: C

[Ref. Page: 173]

Q17. Which operator adds summary row for every combination of groups specified?

- A. ROLLUP
- B. CUBE
- C. GROUPING SETS
- D. OVER

Answer: B

Q18. With GROUPING SETS, how can you add summary row that summarizes the entire result set?

- A. Within parentheses after the GROUPING SETS operator, add a * symbol
- B. Within parentheses after the GROUPING SETS operator, add an empty set of parentheses
- C. After the GROUPING SETS operator, add an empty set of parentheses
- D. You need not do anything; GROUPING SETS add a summary row by default

Answer: B